

Award Nomination App

Technical Executive Overview

The Award Nomination App is a tenant-configurable employee recognition and award governance platform. It combines workflow automation, enterprise identity, tenant-aware configuration, fraud and integrity screening, analytics, notification orchestration, and payroll coordination.

The system is designed as an Azure-native SaaS application that can support multiple tenants, configurable award rules, role-based experiences, asynchronous processing, and operational telemetry.

Purpose

The platform lets employees nominate colleagues for monetary awards, routes nominations through manager and HRBP review, monitors integrity risk, sends workflow notifications, produces analytics, and connects approved awards to payroll processing.

The codebase represents more than a workflow prototype. It includes frontend, backend, worker, analytics, payroll, database, observability, and infrastructure components that form a deployable product foundation.

Core Platform Capabilities

- React/Vite frontend with Microsoft Entra ID sign-in, tenant branding, locale, currency, nomination workflows, approvals, HRBP review, analytics, payroll lookup, and admin experiences.
- FastAPI backend with tenant-aware authentication, role-aware APIs, nomination lifecycle endpoints, analytics endpoints, audit logging, certificate generation, and AI analytics support.
- Azure SQL persistence with schema migration support and tenant-scoped data access patterns.
- Azure Service Bus event routing for asynchronous fraud screening, notifications, HRBP escalation, payout events, and payroll outcomes.
- Integrity-check worker for description quality, semantic checks, ML fraud scoring, and routing decisions.
- Auxiliary worker for notification delivery, idempotent event handling, HRBP alerts, payout support, payroll result handling, and SLA alerts.
- Payroll broker with provider routing and OAuth/webhook patterns for external payroll systems.
- Weekly fraud analytics job for graph-pattern detection, model retraining, holiday sync, and forecasting.
- Terraform and GitHub Actions assets for Azure deployment.
- Application Insights, OpenTelemetry, structured logs, KQL assets, health checks, and dashboards for observability.

Business Problem

Employee recognition programs can become inconsistent, opaque, or vulnerable to abuse when nominations are handled through informal email chains, spreadsheets, or basic approval tools.

Common problems include:

- Favoritism or concentrated awards among a small group.
- Duplicate, copy-paste, or low-effort nominations.
- Reciprocal nominations and nomination rings.

- Awards exceeding tenant or policy limits.
- Weak audit trails for approvals, rejections, payouts, and administrative impersonation.
- Limited executive visibility into spending, trends, review load, risk, and fairness.
- Manual effort to connect approved awards to payroll.

Product Workflow

The platform implements a structured award lifecycle:

1. A user signs in with an organization account.
2. The frontend loads tenant branding, currency, locale, award limits, and award categories.
3. The user submits a nomination for another employee.
4. The backend validates tenant isolation, award amount, category, manager, beneficiary, and description rules.
5. A Service Bus event sends the nomination to the integrity-check worker.
6. The worker runs description quality checks and ML fraud scoring.
7. Clean nominations route to manager approval.
8. Flagged nominations route to HRBP review.
9. Approved nominations trigger notifications, certificate generation, analytics updates, and payroll submission.
10. Admins review analytics, fraud alerts, graph integrity findings, forecasts, logs, and AI-generated investigation support.

Differentiators

- Tenant-aware SaaS architecture with configurable branding, domain, locale, currency, award categories, award ranges, and governance settings.
- Fraud-aware workflow with integrity screening built into the nomination lifecycle before manager approval.
- Graph analytics to detect rings, super nominators, recognition deserts, approver affinity, copy-paste patterns, transactional language, and hidden candidates.
- Human-in-the-loop governance through HRBP review for flagged nominations.
- Operational maturity through telemetry, structured logs, KQL assets, dashboards, health checks, and idempotent event processing.
- Payroll-ready architecture through a payroll broker and provider abstraction.
- Executive analytics across spend, recipients, fraud alerts, approvals, diversity indicators, categories, forecasts, and integrity findings.
- AI analytics workspace for deeper investigation and administrative insight.

Strategic Direction

The Award Nomination App can stand alone as an employee recognition governance product. It also establishes a strong first domain for a broader Integrity Sentinel platform.

The natural expansion path is:

- From award nomination integrity to cross-domain integrity analytics.
- From fraud alerts to investigation workspaces.
- From HR workflows to enterprise risk signals.

- From tenant-specific rules to tenant-specific models and governance controls.
- From isolated recognition payouts to broader HR, payroll, and compliance integration.

Executive Recommendation

Position the platform as an AI-enabled recognition governance product for organizations that need more than lightweight nominations.

The strongest enterprise story is the combination of:

- Workflow configuration.
- Tenant isolation.
- Auditability.
- AI integrity screening.
- Executive analytics.
- Payroll integration.
- Azure-native operational readiness.